

Group Github link: https://github.com/soberox/ITSC-3162-Final_project.git

Kaggle Dataset: <https://www.kaggle.com/datasets/mukeshmanral/fifa-data-for-eda-and-stats>

Poster link:

https://www.canva.com/design/DAG6MG2gw38/LwSsCiXD1MjDeSLVGt88xA/edit?utm_content=DAG6MG2gw38&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

Written Report: Nahuel Nieves - Linear Regression

My full report is written within my Jupyter notebook that is within this GitHub, my file is labeled "FinalProject". As such, I will be posting the GitHub link and providing a summary of my report.

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The main focus of our project was attempting to figure out the worth of soccer players through a FIFA dataset. Utilizing the physical traits, their overall wealth, and general information provided through the dataset to answer our question. I had utilized Linear Regression with a Jupyter notebook, taking advantage of Scikit-Learning and some modeling software such as Matplotlib, Numpy, and Seaborn to go through and iterate the information. Transforming it into a workable heatmap that would assist people in seeing what these values were and what they meant. I primarily focused on the physical aspects of each player, honing the Scikit-Learning model towards those physical attributes.

Things progressed rather quickly as I started my code, which once again is within the GitHub labeled as "FinalProject". My first concern quickly arose as my accuracy was only around 43%. Of course this would have been a concern if it was not for the fact that each player had a vastly different set of skills compared to one another, which for a team based game, is expected. Each player is not supposed to be an all star player, they have to have specific skills that the team is looking towards and is able to play well. As such, when I created my heatmap, this was reflected heavily as the values did not show many that certainly outweighed others. Apologies to the goalies but unless skills were superb, they were likely not getting selected for most roles.

In terms of an impact, it certainly helped my own views regarding the sport as I always viewed it as biased. This is due to the fact that a lot of people tend to prefer people who were more flamboyant, and their skills seemed questionable at times but when compared to other players within the same bracket and skill set, it was difficult to see who would actually win. As such, given that these players must also be television personalities, it is not hard to see why it is so difficult to find a talent that can do both well.

Written report by Jonathan Corll - XGBoost Regression

The main goal of this project was to learn the best methods to value and analyze soccer players using a dataset of all players included in the FIFA 21 video game. Using this dataset of all soccer players, I sought to answer the question of whether we can predict a players' value based on their demographic profile and their attributes. The goal for answering this question was to learn valid methods for analyzing real life players' values by getting some practice on subjective, video game data. The dataset included an overall, the potential overall when considering player development, a value and a wage variable, and then stats for all aspects of the game that look at their offensive, defensive, and game abilities.

My pre-processing included eliminating all goalkeepers and goalkeeping stats, as that would have skewed my data and brought down all the predicted values for the field players, so I deemed that an acceptable move. I also had to change all the variables that were strings to numeric values, either through conversions (height and weight) or different encoding techniques (nationality, club, position, etc.). Lastly, I had to deal with null values, which I dropped irrelevant columns, and then used appropriate techniques to fill in certain columns where I could eliminate the missing values without ruining the validity of my data.

To evaluate my data, I used statistical scores like Mean Absolute Error, Root Mean Squared Error, and an R-squared value. My R-squared value came back around 94%, which means that the model was quite accurate and did not have much variance, meaning it had little unexplainable error. For my two mean error scores, my Root Mean Squared Error was higher than the Mean Absolute Error, which meant that the larger values in the dataset were, on average, further off than the values that sat around the average range. Meaning that for the most valuable players, my model was more off than it was for the majority of values in the median range. And my Mean Absolute Error was only around 400K Euros, which when compared to the sum of most values being in the multi-millions, that is a small error. Taking all this into account, my model performed well and did an excellent job in predicting the true value of the players based on their statistics.

I accomplished the goal of my question, as I found a way to predict the value of a player based on their stats. I learned that looking at and analyzing a player's on-field performance goes a long way in determining how valuable they will be, however, it cannot just be on their offensive output.. Players who excel at defense and possession skills should be valued just as much as players who can score many goals and generate a large number of assists. Controlling all levels of a field is important and players should be highly valued even if their box score stats are not the highest. One step I would like to do in the future is to look at a feature importance chart for my model. That would show me what features and skills are the most important in determining a player's value, and that would give great insight to any coach or scout.

However, even without the feature importance graph, I believe this project has a positive impact for coaching staffs and scouts around the world in soccer. I believe this gives them potentially new tactics for player transfers and opens their eyes to the impact that analytics can have in team construction. It could also provide a headstart on automating parts of the scouting process, giving scouts more time for their most important tasks and can automatically flag players who need to be eliminated from transfer consideration, or flag players who need to be pursued or scouted further. Overall, this project should have a positive impact on the soccer community and how we value players and determine just how much they are worth on the transfer market.

Written by Amariah Robinson 😊 - Agglomerative Hierarchical Clustering

The main goal of this project was to gain insight about different methods of analyzing the dataset and see what we can conclude from the analyses. The dataset was from Kaggle and it was from the FIFA 21 EA Sports game.

The dataset itself contains many features that can describe many attributes of a player such as ball_control, height, club_name, etc. The dataset contains almost 20 thousand rows of information, along with over 50 columns. The key features that I wanted to focus on were Value and Wage of a player.

To get this dataset prepped and ready for modeling, we first had to fill in any missing values that we were missing from the dataset. Next was to make sure that the model would be able to understand the weight of the values given. So changing currency values from, for example, €110K to €110,000 was a step that needed to be done! Now that that has been completed, we can start to create our model. The goal for this was to compare the value and wages of players to other players and see if we can cluster players to see if some are more similar than others. I decided to use Agglomerative Hierarchical Clustering (AHC) because it would group together players with similar stats and continue until a group of players, or individual players, reach the top of the food chain, representing the most valued and paid player. These players would then be grouped together overall connecting the subgroups of players, creating a dendrogram. Now mind you, there are over 18 thousand rows of information right? So think about clustering together 18 thousand rows into just around 10 groups. Insanity. But, what I found out about dendrograms is that you can clean them up and look so much neater. So, instead of showing every leaf of subgroups, I was able to just show the number of groups within the one cluster. Does this explicitly explain who is in said subgroup? No. But, this shows that all players were able to fit into subgroups and shows their hierarchical status. Now from the dataset, we can see that L. Messi and C. Ronaldo are leading players. This gives us an insight on what the dendrogram is saying about the players, showing that they are top of the food chain when it comes to their wages and values.

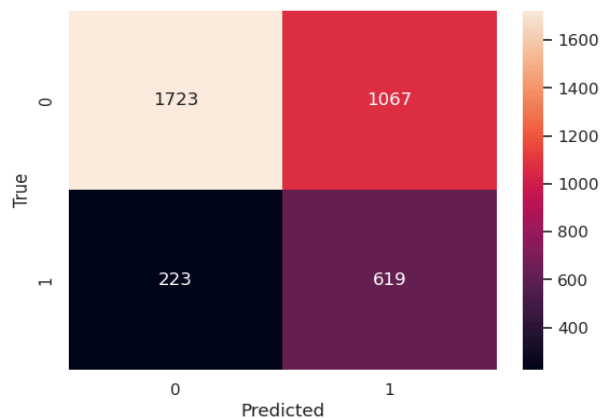
Some insights gained from completing this model is that looks can be deceiving. I didn't know what a dendrogram was before beginning this project and now I feel more confident in the ins and outs of it. Through the use of the dendrogram, I was able to answer the question of, can players be clustered based on their wage and value. They're more similar than we may think! Some things that could be improved would be to include some way to incorporate metrics and statistics into the mix, showing that yeah, this model isn't just some pretty photo. Things like using a silhouette score or what is called a Cophenetic Correlation Coefficient (CCC), which shows how well the distances between branches were compared to the actual dataset and distances between values for each player. Basically maintaining the integrity of the data while transforming it into a visual representation.

Thinking back to how we used someone's subjective thinking to create models and provide different types of evidence goes to show how easy it is to transform bad data into what can look like a million bucks. This brings up the ethical side of data science overall and the ability to produce a different perspective than what may be given originally. Now imagine that on a larger scale, with billion dollar companies who could be fabricating data and misrepresenting it, all for a check. Makes you think about data and how sensitive it is, no matter who is the beholder!

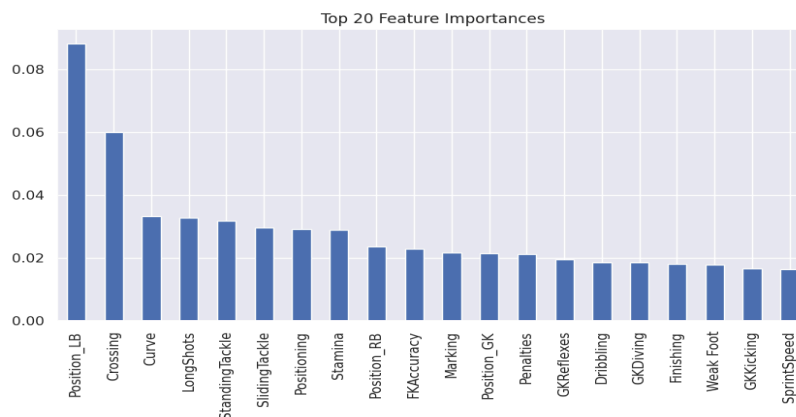
Written by Yousef Eddin - Random Forest Classification

The main goal was to make some sort of classification model using the Fifa Dataset. Unfortunately, the majority of the columns were numerical and or not related to the players performance. So I decided, ultimately, to pick whether the player was left or right footed.

My approach goes as follows: Importing the dataset, cleaning the dataset, doing a quick exploratory data analysis, preparing the dataset for modeling, modeling, and finally having our conclusion. During the cleaning section of our data, I found that all the columns that has to do with money have the "\$", "€", "K", or "M". I had to create a function to search the string and remove all the currency and decimal units and turn them into their actual numbers. Converting a string to a float. The second problem I learned during the preparation for modeling was that left footed and right footed players are completely off balance. More than double the data are right footed than left footed. This is a key factor when modeling. The model I used was RandomForestClassifier. This is because RandomForest is extremely resistant to overfitting and can easily find which columns are most important. While creating the random forest model, I used `class_weight = 'balanced'`. This makes it so the two datapoints are balanced instead of insanely unbalanced. After that, I measured the dataset using precision, and accuracy. The following is a confusion matrix of the data.



Given that this is synthetic data from a game, I'm pretty happy with the results! The fact that the left footed players are about only 20% of all players, the fact that we're getting a lot of false negatives is actually a pretty good sign.



These are the top 20 important features. The benefit of random forest is that we can see what's most important and these are the most important things.

Written Report Jean Andrade - Recommendation System

This project focuses on developing a recommendation system for football club owners who must make data driven decisions when evaluating players. Football's global popularity and the enormous diversity of player skillsets make it difficult for clubs to manually compare athletes across leagues, positions, and attributes. The goal of this project is to build a reproducible, quantitative system that can recommend players similar to a chosen athlete based on real performance and skill indicators. By leveraging structured player statistics, the project seeks to answer practical questions such as: Which players resemble a given player the most? What factors influence similarity? And how can owners use statistics to guide scouting and recruitment?

The dataset used for this project comes from a FIFA player statistics collection publicly available on Kaggle. It contains detailed information about thousands of players, including numeric features such as overall rating, potential, sprint speed, dribbling, and defensive attributes, as well as categorical metadata such as nationality, club, and preferred foot. After loading the data, exploratory inspection revealed a rich and sufficiently large dataset for building a meaningful recommendation engine. The project further explored the data through summary statistics, inspection of data types, and plotting correlation matrices, which all helped clarify which features were relevant for modeling and which were redundant or noisy.

To prepare the dataset for analysis, several preprocessing steps were required. Irrelevant or noninformative columns such as ID values, loan information, and contract expiration date were removed because they offered no predictive value in similarity based recommendations.

Categorical variables were encoded using dummy variables so they could be included in vector based similarity calculations. The numeric features were standardized using a scaling method to ensure that attributes measured in different ranges contributed equally to the model. These steps produced a clean feature matrix suitable for modeling, and they reduced bias from raw, unnormalized measurements. The workflow also involved experimenting with feature selection and removing highly correlated or low variance features to improve the quality of the similarity output.

The recommendation engine itself uses a straightforward yet effective vector space approach. Once all player attributes are processed into numeric feature vectors, the system locates a specific player in the dataset, extracts that player's feature vector, computes distances or similarities between that vector and all others, and returns the nearest neighbors. This method produced consistent, interpretable results: recommended players tended to share similar roles, physical profiles, and overall skill ratings with the target player. Additional experimentation involved selecting alternative feature subsets and testing whether clustering by overall score produced meaningful separations, which helped validate that the model's structure aligned with real-world player categorization.

Evaluating the recommendation system involved examining whether the returned players made intuitive sense when compared to the target player. Because the goal here is not classification accuracy but qualitative similarity, evaluation focused on consistency and interpretability: whether similar positions cluster together, whether players with comparable skill distributions appear near one another, and whether removing or adding features changes the stability of recommendations. Through this process, the project successfully demonstrated that

standardized numeric attributes combined with encoded categorical values form a solid foundation for a recommendation system, and the model reliably identifies close statistical matches.

This project highlights the usefulness of data driven scouting tools but also raises important considerations about impact. A system like this could support more equitable recruiting by helping smaller clubs access analytical insights typically reserved for wealthier organizations; however, it could also reinforce existing biases embedded in historical data or overemphasize quantifiable attributes at the expense of contextual, human level assessment. These models must therefore be used responsibly, acknowledging that real players are more than the numbers representing them. Overall, the project demonstrates the power of reproducible analytics, reinforces key data science concepts from the course, and provides a foundation for future work such as integrating positional clustering, weighting certain attributes more heavily, or incorporating time-series performance data to strengthen predictive insights.